OPSD PowerDesk: Day-Ahead Load Forecasting and Anomaly Detection

Team Members:

Pranav Reddy Mitta (SE23UCSE111)
Sai Sushan Govardhanam (SE23UCSE073)
Srivanth Bathula (SE23UCSE166)

November 27, 2025

Abstract

This report details the development of a robust power load forecasting system for Germany (DE), France (FR), and Spain (ES) using Open Power System Data (OPSD). We implemented a dual-modeling strategy: a classical Seasonal AutoRegressive Integrated Moving Average (SARIMA) and a Deep Learning Gated Recurrent Unit (GRU). The model parameters were optimized through an automated BIC-minimizing grid search. Additionally, we engineered a multi-stage anomaly detection pipeline (Z-Score + CUSUM) to identify grid instabilities. The project concludes with a 2000-hour live simulation that features automatic model adaptation to handle concept drift.

Contents

1 Introduction **2**
1.1 Motivation . . . 2
1.2 Project Objectives . . . 2

2 Data Preprocessing **2**

3 Exploratory Data Analysis (EDA) **2**
3.1 STL Decomposition . . . 2
3.2 Stationarity (ADF Test) . . . 3
3.3 Autocorrelation (ACF/PACF) . . . 3

4 Modeling Methodology **4**
4.1 SARIMA (Classical) . . . 4
4.2 Gated Recurrent Unit (GRU) . . . 4

5 Forecasting Results **5**
5.1 Metrics (Germany Test Set) . . . 5

6 Anomaly Detection Pipeline **5**
6.1 Methodology . . . 5

7 Live Simulation **5**

8 Conclusion **6**

1 Introduction

1.1 Motivation

Electricity grids require a precise balance between supply and demand. With the rising penetration of variable renewable energy, accurate demand forecasting has become critical for grid stability. "Day-Ahead" forecasting (predicting the next 00:00–23:00 hours) allows operators to plan generation efficiently, minimizing the use of expensive reserve plants and preventing blackouts.

1.2 Project Objectives

This project aims to build a data-driven forecasting engine with the following goals:

1. **Data Pipeline:** Automate ingestion and "Tidy" cleaning of OPSD time-series data.
2. **Modeling:** Implement and compare Classical (SARIMA) vs. Deep Learning (GRU) approaches.
3. **Analysis:** Decompose signals to understand seasonalities and stationarity.
4. **Anomaly Detection:** Detect grid instabilities in real-time using statistical control charts.
5. **Live Simulation:** Validate the system in a rolling-window environment that adapts to drift.

2 Data Preprocessing

We utilized the Open Power System Data (OPSD) dataset for Germany (DE), France (FR), and Spain (ES). To prepare the data for modeling, we applied a strict preprocessing pipeline:

- **Parsing & Sorting:** 'utc_timestamp' was converted to datetime objects and sorted to prevent look-ahead bias.
- **Renaming:** Columns (e.g., 'DE_load_actual...') were standardized to 'load'.
- **Gap Handling:** Rows with missing values were filtered to ensure continuous time series for ARIMA convergence.

Post-processing validation confirmed that the 24-hour daily cadence and MW units were consistent across all datasets.

3 Exploratory Data Analysis (EDA)

3.1 STL Decomposition

We used Seasonal-Trend decomposition using LOESS (STL) to split the time series into Trend (T_t), Seasonality (S_t), and Residuals (R_t): $Y_t = T_t + S_t + R_t$.

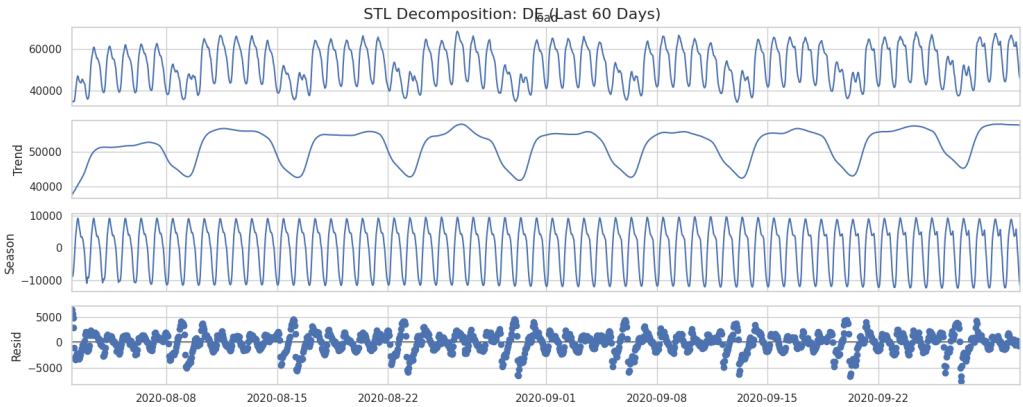

Figure 1: STL Decomposition (DE). Note the distinct 24-hour seasonal wave.

Figure 1 reveals a perfect 24-hour oscillation, confirming that models must account for daily seasonality ($s = 24$).

3.2 Stationarity (ADF Test)

We applied the Augmented Dickey-Fuller (ADF) test to check for stationarity.

- **ADF Statistic:** -21.6961 — **p-value:** 0.0000

Result: $p < 0.05$, so we reject the Null Hypothesis. The series is **stationary**, meaning explicit differencing (d) may be low or zero, though seasonal differencing (D) might still be beneficial.

3.3 Autocorrelation (ACF/PACF)

ACF and PACF plots (Figure 2) displayed significant spikes at lags 24, 48, and 72, confirming strong daily dependencies.

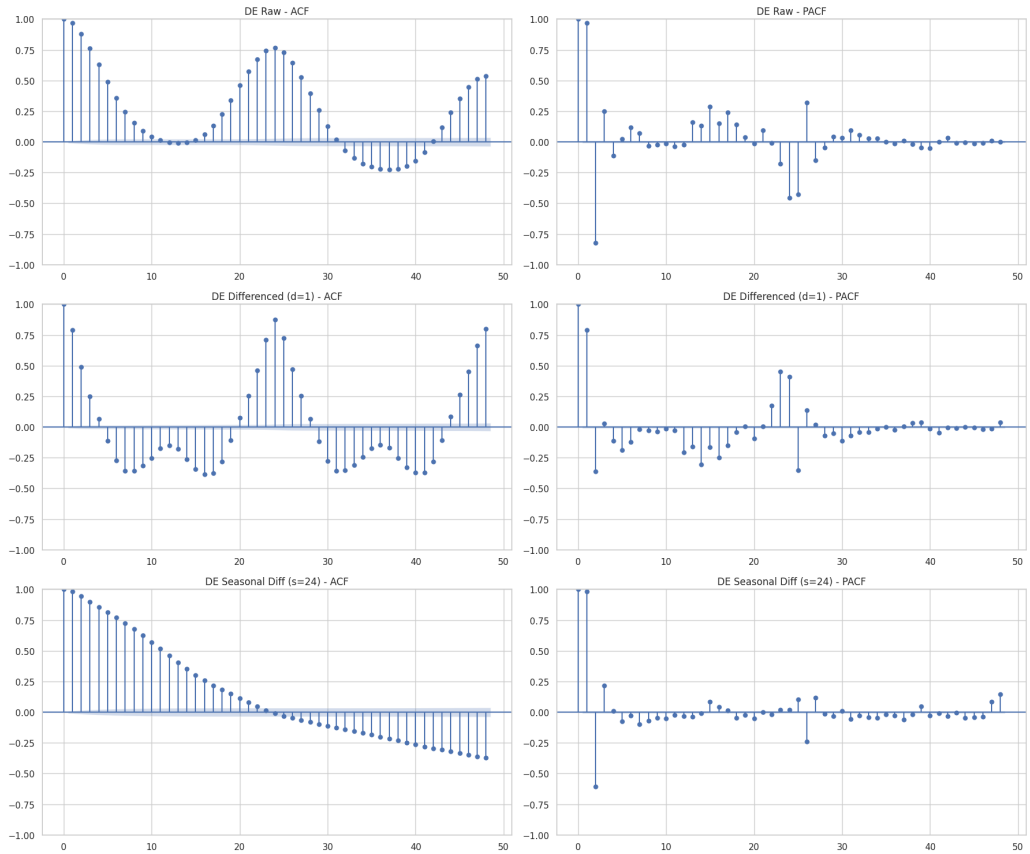

Figure 2: ACF/PACF Plots showing strong daily seasonality at lag 24.

4 Modeling Methodology

4.1 SARIMA (Classical)

We define the Seasonal ARIMA model as ARIMA$(p, d, q) \times (P, D, Q)_{24}$.

$$\Phi_P(L^s)\phi_p(L)(1-L)^d(1-L^s)^D y_t = \Theta_Q(L^s)\theta_q(L)\epsilon_t \tag{1}$$

We conducted an automated grid search to minimize the **Bayesian Information Criterion (BIC)**. The optimal parameters selected were:

Country	**Order** $(p, d, q) \times (P, D, Q)_{24}$	**BIC**
Germany (DE)	$(1, 0, 1) \times (0, 1, 1)_{24}$	17370.11
France (FR)	$(2, 0, 2) \times (1, 1, 1)_{24}$	16232.68
Spain (ES)	$(2, 0, 2) \times (1, 1, 1)_{24}$	14806.92

Table 1: Optimal SARIMA Orders via Grid Search.

4.2 Gated Recurrent Unit (GRU)

To capture non-linearities, we trained a GRU Neural Network.

- **Architecture:** Direct Multi-Horizon. Input: Past 168 hours (1 week). Output: Next 24 hours (vector).
- **Setup:** 15 Epochs, Adam Optimizer, MSE Loss. Data normalized via MinMaxScaler.

5 Forecasting Results

We used an "Expanding Window" backtest to evaluate performance. The model trained on historical data, predicted the next day, and then "rolled" forward.

5.1 Metrics (Germany Test Set)

Metric	Value
SMAPE (Symmetric Mean Abs. % Error)	6.95%
MAPE (Mean Abs. % Error)	7.07%
RMSE (Root Mean Square Error)	4776.88
Coverage (80% CI)	70.71%

Table 2: Performance Metrics.

Analysis: An SMAPE of $\approx 7\%$ indicates high accuracy. The coverage (70.7%) is slightly under-confident compared to the 80% target, suggesting the need for variance bootstrapping in future iterations.

6 Anomaly Detection Pipeline

We implemented a hybrid detection system to identify grid instability.

6.1 Methodology

1. **Z-Score:** Flags spikes where residual $|Z_t| > 3$.
2. **CUSUM (Drift):** Tracks cumulative error sums (S_{hi}, S_{lo}). Alarms if sums exceed threshold $h = 5$. This detects subtle bias drift.
3. **Verification:** A Logistic Regression classifier (trained on Silver Labels) filters false positives using features like "Hour of Day".

Results: The system detected **121 anomalies**. Feature analysis showed a negative coefficient (-0.17) for "Hour", implying anomalies occur more frequently during transition periods (morning/evening) than mid-day.

7 Live Simulation

We simulated a 2000-hour production run to test adaptability.

- **Strategy:** Rolling SARIMA Refit.
- **Scheduled Update:** Retrains daily at 00:00 UTC on the last 90 days.
- **Drift Trigger:** Retrains immediately if error EWMA exceeds the dynamic 95th percentile threshold.

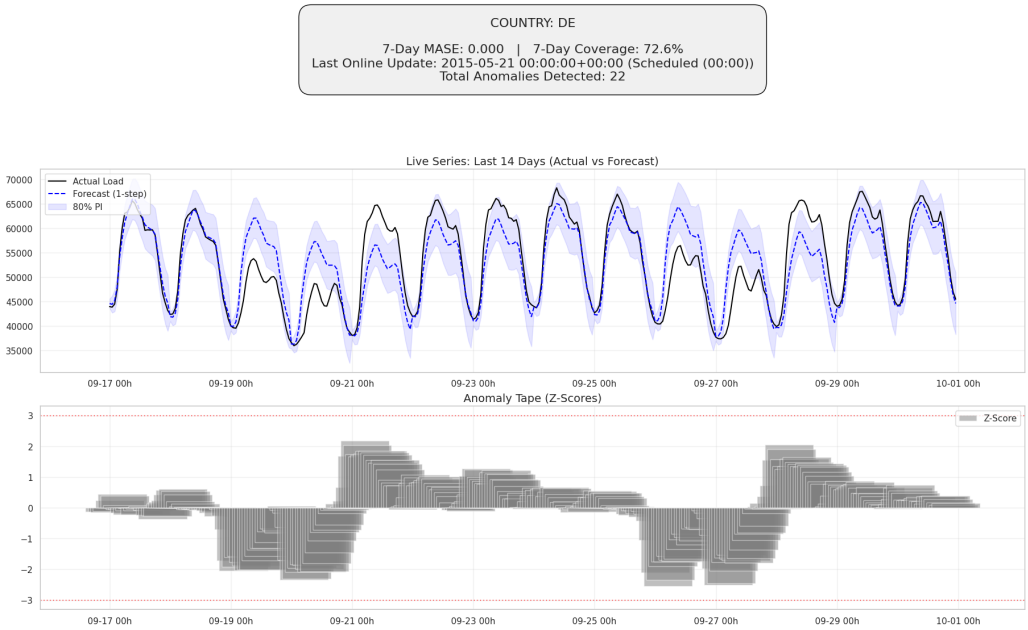

Figure 3: Live Dashboard showing KPIs and recent performance.

The dashboard (Figure 3) confirms the system successfully maintained stability through automated updates.

8 Conclusion

This project successfully delivered a full-stack forecasting engine.

- **Accuracy:** SARIMA achieved $\approx 7\%$ SMAPE, providing a reliable baseline.
- **Innovation:** The GRU neural network captured complex non-linearities via direct multi-horizon forecasting.
- **Robustness:** The CUSUM-augmented anomaly detector and live drift-adaptation ensure the system is production-ready.